

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)	(i)		The 3 readings are similar / the 3 readings are repeatable			1	1	0	1
		(ii)		$\frac{2.82}{3} = 0.94$ (1)		1		1	1	1
	(b)	(i)		Ticks in boxes 2, 4 and 6 • Cake case 1 and 2 have identical weight (1) • At terminal speed, both cake cases experience identical values of air resistance (1) • At terminal speed, both cake cases have zero acceleration (1)		3		3		3
		(ii)		Terminal speed of CK1 is 1.6 and when doubled is 3.2 m/s (1) which isn't the same as 2.3 m/s for CK2 (1) [so prediction incorrect.] Alternative: Terminal speed of CK2 is 2.3 m/s and when halved is 1.15 m/s (1) which isn't the same as 1.6 m/s for CK1 (1) [so prediction incorrect.] Alternative: It is not doubled (1) It increases by 0.7 m/s (1) [so prediction incorrect.]			2	2	2	2
	(c)			Ticks in boxes 2, 4 and 6 • The terminal speed after the parachute is opened is $\frac{1}{10}$th of the terminal speed before the parachute is opened (1) • The parachute is opened 30 s after the skydiver leaves the helicopter (1) • At point A the skydiver stops accelerating (1)	1	1		3	1	
				Question 11 total	1	6	3	10	4	7